Supplementary Material 1

Programme-Dependent Bias and Uncertainty in Mass-Loss-Based Estimates of Retained Ultimate Strength of Artificially Corroded Reinforcing Steel

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Supplementary Methods S1. Analysis sequence and verification

S1.1 Analysis sequence. The analysis scripts run in a fixed sequence: data verification; LOPO and sensitivity runs; random-record diagnostic; interval and novelty diagnostics; programme bootstrap; tables; figures; and a final re-implementation check.

S1.2 Data-integrity gate. The data-integrity gate verifies the following and halts on any mismatch: the SHA-256 checksums of the original workbook, the user's manual and the frozen cohort file; the cohort counts (707 tests, 11 programmes, 646 ultimate-strain values, no averaged rows, deformed bars only); and cell-level agreement of mass loss, retained strength and specimen names with the original workbook.

S1.3 Re-implementation check. The final check re-implements the M1 and M2 LOPO pipelines separately from the main analysis code and reproduces all stored predictions (maximum absolute difference $< 10^{-8}$). It also recomputes the metrics, bootstrap limits and interval half-widths from the stored predictions and asserts that the held-out programme never enters training. A complete rerun from the frozen data reproduced all numerical outputs byte for byte.

S1.4 Cohort-derivation counts (Fig. S4). The counts in Fig. S4 are computed, without modification of any record, from the ingested source database (1,387 rows) and the frozen 707-test cohort file: the number of tests and of distinct programmes for each corrosion-method code; the number of records flagged by the keyword rule ("average" in the source Extra Notes field) overall (62, of which 54 are reported averages of several tests and 8 are individual tests whose notes refer to an average of another quantity) and within the impressed-current subset (0), so that none of these records is included in the 707-test cohort; and the number of impressed-current tests removed by criteria (ii)–(vi) (0). The load-history counts in Table S9 are taken from the same source file.

S1.5 Software. All analyses used Python 3.13.17 with NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, scikit-learn 1.9.1 and Matplotlib 3.11.2. The code that implements the analysis and the figures is available from the corresponding author upon reasonable request.

Supplementary Methods S2. Model forms, metrics and interval construction

M1 (ridge regression) is linear in standardised mass loss $z = (m - \bar{m}_{\mathrm{tr}})/s_{\mathrm{tr}}$, with training-only standardisation, an unpenalised intercept and penalty $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ selected by inner grouped four-fold validation (criterion: inner programme-macro MAE; ties to the first candidate):

$$\begin{aligned} \hat{R}_{\mathrm{M1}} &= \hat{\beta}_0 + \hat{\beta}_1 z, \\ (\hat{\beta}_0, \hat{\beta}_1) &= \arg\min_{\beta_0,\beta_1} \sum_{i\in\mathrm{tr}} (R_i - \beta_0 - \beta_1 z_i)^2 + \alpha\beta_1^2. \end{aligned} \tag{S1}$$

M2 (weighted isotonic regression) fits a non-increasing step function of mass loss with weights normalised to unit mean, so that each training programme contributes equally; predictions outside the training range are clipped to the boundary values:

$$\begin{aligned} \hat{f} &= \arg\min_{f \text{ non-increasing}} \sum_{i\in\mathrm{tr}} w_i \left(R_i - f(m_i)\right)^2, \\ w_i &\propto \frac{1}{n_{g(i)}}, \end{aligned} \tag{S2}$$

M3 (secondary) is a histogram-based gradient-boosting regressor with inputs (m, d), non-increasing in m, early stopping disabled, fixed seed and one of three predeclared configurations (Table S12). *Per-programme metrics* (error $e_i = \hat{R}_i - R_i$):

$$\begin{aligned} \mathrm{MAE}_g &= \frac{1}{n_g} \sum_{i \in g} |e_i|, \\ \mathrm{Bias}_g &= \frac{1}{n_g} \sum_{i \in g} e_i, \\ \mathrm{OR}_g &= \frac{100}{n_g} \sum_{i \in g} \mathbb{1}(e_i > 0), \\ \mathrm{PO}_g &= \frac{1}{n_g} \sum_{i \in g} \max(e_i, 0), \end{aligned} \tag{S3}$$

the MAE, signed bias, overprediction rate (%) and mean positive overprediction (pp); programme-macro values average over the 11 programmes, specimen-micro values pool all 707 specimens. *Prediction intervals*: cross-fitted residuals r_j of all training specimens (inner grouped folds; M0 residuals computed directly) receive weights $\omega_j = 1/(G_{\mathrm{tr}}\, n_{g(j)})$, $G_{\mathrm{tr}} = 10$, and the half-width is the weighted 0.90 quantile of $|r|$, without interpolation or finite-sample inflation:

$$\begin{aligned} q &= \min\left\{ |r|_{(k)} : \sum_{j \le k} \tilde{\omega}_{(j)} \ge 0.90 \right\}, \\ \mathrm{PI} &= [\hat{R} - q,\ \hat{R} + q], \end{aligned} \tag{S4}$$

Mass-loss-domain novelty: the primary measure is the extrapolation distance outside the training support, and the secondary measure d_{kNN} the mean distance to the ten nearest training mass-loss values, both scaled by the training interquartile range:

$$d_{\mathrm{ext}} = \frac{\max(0,\ m_{\min} - m,\ m - m_{\max})}{\mathrm{IQR}_{\mathrm{tr}}}. \tag{S5}$$

Errors, bias, overprediction rate and coverage are also summarised in predeclared mass-loss bands (<10, 10–20, 20–30, 30–50 and ≥50 %; Table S5). The record-random diagnostic used five-fold shuffled K-fold cross-validation over specimens (seed 20261027) with the same nested grouped tuning.

Supplementary tables

Tables S2 (complete per-programme metrics for M0–M3) and S8 (inner-tuning selections and scores) are provided only as CSV files (Supplementary Material 2) because of their size. All tables are also supplied as CSV files with identical values in Supplementary Material 2.

Table S1. Within-programme descriptive relationships between retained strength and mass loss, and overlap of mass-loss ranges (descriptive only).

Programme	n	Mass loss IQR (%)	Retained strength IQR (%)	OLS slope (pp/%)	Intercept (%)	Spearman rho	Covering programmes (median)	Share of cohort (%)
Almusallam (2001)	42	4.2-28.1	51.6-95.9	-1.10	96.3	-0.89	5	5.9
Cobo et al. (2011)	85	6.7-12.2	82.6-91.7	-1.37	99.3	-0.82	8	12.0
Fernandez et al. (2015)	35	9.8-16.9	67.7-79.1	-1.91	99.0	-0.83	9	5.0
Imperatore et al. (2017)	65	10.8-23.0	67.1-86.1	-1.41	99.1	-0.82	8	9.2
Kashani et al. (2013b)	5	15.5-16.1	64.9-73.3	-1.48	91.7	-0.10	10	0.7
Meda et al. (2014)	22	16.1-20.9	62.3-71.2	-2.00	102.0	-0.71	8	3.1
Ou et al. (2016)	29	8.4-15.3	79.0-90.0	-1.10	99.0	-0.79	8	4.1
Tariq and Bhargava (2021)	120	9.6-30.5	53.2-82.7	-1.26	94.9	-0.95	6	17.0
Matthews et al. (2023)*	272	4.9-9.0	89.8-94.6	-1.06	99.6	-0.91	7	38.5
Vanama and Ramakrishnan (2020)	12	17.8-35.3	57.0-76.3	-1.01	95.4	-0.96	6.5	1.7
Zhang et al. (2012)	20	9.6-26.0	72.2-88.2	-1.07	97.6	-0.80	7	2.8

Table S3. Paired programme-level comparisons (A − B; negative values favour A) with 95 % programme-bootstrap intervals (5000 resamples; descriptive).

Quantity	M1-M0	M2-M0	M2-M1	M3-M2
Macro MAE diff (pp) [95% PB]	-1.58 [-3.61, 0.41]	-2.32 [-4.81, 0.03]	-0.74 [-1.68, 0.22]	0.33 [-0.13, 0.85]
Median programme diff (pp)	-1.6	-1.65	-1.49	0.05

Quantity	M1-M0	M2-M0	M2-M1	M3-M2
Programmes A lower / B lower	6 / 5	7 / 4	6 / 5	5 / 6
Leave-one-programme-out range of macro diff (pp)	-2.13 to -1.11	-2.97 to -1.57	-1.01 to -0.47	0.11 to 0.44
Macro bias diff (pp) [95% PB]	-6.36 [-7.34, -5.46]	-7.36 [-8.34, -6.30]	-1.01 [-2.13, 0.20]	-0.12 [-1.07, 0.80]
Macro overpred. rate diff (pct-pts) [95% PB]	-32.7 [-50.8, -16.7]	-38.8 [-52.7, -25.6]	-6.1 [-15.1, 1.1]	-1.9 [-8.8, 5.3]

Table S4. Record-random five-fold diagnostic versus LOPO, with identical metric definitions.

Model	Random-record macro MAE (pp)	LOPO macro MAE (pp)	LOPO - random, macro (pp) [95% PB]	Programmes LOPO worse / better / equal	Random-record micro MAE (pp)	LOPO micro MAE (pp)	Random-record micro bias (pp)	LOPO micro bias (pp)
M0	8.34	8.34	0.00 [0.00, 0.00]	0 / 0 / 11	5.86	5.86	5.17	5.17
M1	6.36	6.76	0.40 [0.12, 0.82]	8 / 3 / 0	4.71	5.68	-0.07	-0.90
M2	5.53	6.02	0.49 [0.23, 0.75]	9 / 2 / 0	4.35	4.83	-0.61	-0.85
M3 (secondary)	5.37	6.34	0.97 [0.60, 1.37]	11 / 0 / 0	3.83	5.04	0.04	-1.23

Table S5. LOPO errors, bias, overprediction rate and 90 % interval coverage by mass-loss band.

Model	Mass-loss band (%)	n	Programmes	PI coverage (%)	MAE (pp)	Bias (pp)	Overpredicted (%)
M0	<10	364	10	99.73	2.49	1.73	73.08
M0	10-20	214	11	93.93	7.94	7.36	87.38
M0	20-30	75	9	77.33	12.3	11.33	85.33
M0	30-50	46	6	86.96	12.34	12.05	97.83
M0	>=50	8	3	87.5	6.01	6.01	100.0
M1	<10	364	10	99.73	4.74	-3.09	20.88
M1	10-20	214	11	94.39	5.82	1.75	55.61
M1	20-30	75	9	85.33	8.08	2.94	57.33
M1	30-50	46	6	93.48	6.4	0.4	52.17
M1	>=50	8	3	25.0	18.44	-15.53	12.5
M2	<10	364	10	99.73	3.39	-1.34	25.82
M2	10-20	214	11	97.2	5.34	-0.97	44.39
M2	20-30	75	9	84.0	8.52	0.32	48.0
M2	30-50	46	6	91.3	7.27	1.64	52.17
M2	>=50	8	3	75.0	7.61	-0.76	37.5

Table S6. Interval half-width, width, calibration sizes and coverage by held-out programme.

Model	Held-out programme	Half-width q (pp)	Calibration programmes	Calibration residuals	Test specimens	Coverage (%)	Width (pp)
M0	Almusallam (2001)	19.26	10.0	665.0	42	83.33	38.51
M0	Cobo et al. (2011)	19.26	10.0	622.0	85	100.0	38.51
M0	Fernandez et al. (2015)	19.26	10.0	672.0	35	82.86	38.51
M0	Imperatore et al. (2017)	19.26	10.0	642.0	65	92.31	38.51
M0	Kashani et al. (2013b)	18.12	10.0	702.0	5	60.0	36.24
M0	Meda et al. (2014)	18.9	10.0	685.0	22	72.73	37.79
M0	Ou et al. (2016)	19.26	10.0	678.0	29	100.0	38.51
M0	Tariq and Bhargava (2021)	19.26	10.0	587.0	120	92.5	38.51
M0	Matthews et al. (2023)*	19.26	10.0	435.0	272	99.26	38.51
M0	Vanama and Ramakrishnan (2020)	19.26	10.0	695.0	12	100.0	38.51
M0	Zhang et al. (2012)	19.26	10.0	687.0	20	95.0	38.51
M1	Almusallam (2001)	14.27	10.0	665.0	42	76.19	28.55
M1	Cobo et al. (2011)	14.77	10.0	622.0	85	100.0	29.54
M1	Fernandez et al. (2015)	14.01	10.0	672.0	35	85.71	28.01
M1	Imperatore et al. (2017)	13.86	10.0	642.0	65	92.31	27.71
M1	Kashani et al. (2013b)	13.49	10.0	702.0	5	60.0	26.99
M1	Meda et al. (2014)	13.75	10.0	685.0	22	77.27	27.49
M1	Ou et al. (2016)	14.12	10.0	678.0	29	100.0	28.25

Model	Held-out programme	Half-width q (pp)	Calibration programmes	Calibration residuals	Test specimens	Coverage (%)	Width (pp)
M1	Tariq and Bhargava (2021)	15.12	10.0	587.0	120	97.5	30.24
M1	Matthews et al. (2023)*	13.32	10.0	435.0	272	99.63	26.64
M1	Vanama and Ramakrishnan (2020)	13.83	10.0	695.0	12	83.33	27.66
M1	Zhang et al. (2012)	14.16	10.0	687.0	20	100.0	28.32
M2	Almusallam (2001)	16.37	10.0	665.0	42	90.48	32.74
M2	Cobo et al. (2011)	17.18	10.0	622.0	85	100.0	34.36
M2	Fernandez et al. (2015)	15.89	10.0	672.0	35	100.0	31.79
M2	Imperatore et al. (2017)	14.37	10.0	642.0	65	89.23	28.74
M2	Kashani et al. (2013b)	13.34	10.0	702.0	5	80.0	26.69
M2	Meda et al. (2014)	14.23	10.0	685.0	22	86.36	28.47
M2	Ou et al. (2016)	13.13	10.0	678.0	29	96.55	26.26
M2	Tariq and Bhargava (2021)	16.54	10.0	587.0	120	99.17	33.07
M2	Matthews et al. (2023)*	15.61	10.0	435.0	272	99.26	31.23
M2	Vanama and Ramakrishnan (2020)	13.88	10.0	695.0	12	100.0	27.76
M2	Zhang et al. (2012)	12.23	10.0	687.0	20	70.0	24.46

Table S7. Prespecified sensitivity analyses.

Analysis	Model	Specimens	Programmes	Macro MAE (pp)	Micro MAE (pp)	RMSE (pp)	Micro bias (pp)	Macro bias (pp)	Macro overpredicted (%)
Primary analysis	M0	707	11	8.34	5.86	8.74	5.17	7.65	85.01
Primary analysis	M1	707	11	6.76	5.68	7.08	-0.9	1.29	52.28
Primary analysis	M2	707	11	6.02	4.83	6.51	-0.85	0.28	46.16
S1: programmes with at least 10 tests	M0	702	10	7.66	5.79	8.66	5.1	6.9	83.52
S1: programmes with at least 10 tests	M1	702	10	6.44	5.63	7.02	-0.88	0.53	47.68
S1: programmes with at least 10 tests	M2	702	10	5.92	4.69	6.44	-0.48	0.07	47.15
S2: embedded specimens only	M0	504	8	8.64	4.26	7.19	3.52	7.85	81.05
S2: embedded specimens only	M1	504	8	7.86	5.69	7.19	-1.67	2.83	57.05
S2: embedded specimens only	M2	504	8	6.9	4.55	6.41	-1.48	0.75	48.29
S3a: excluding R above 100 %	M0	703	11	8.33	5.84	8.69	5.26	7.74	85.52
S3a: excluding R above 100 %	M1	703	11	6.75	5.72	7.03	-0.94	1.28	51.97
S3a: excluding R above 100 %	M2	703	11	5.99	4.85	6.44	-0.91	0.27	46.01
S3b: excluding the 3 highest-mass-loss tests	M0	704	11	8.37	5.87	8.76	5.18	7.66	84.78
S3b: excluding the 3 highest-mass-loss tests	M1	704	11	6.66	5.44	6.77	-0.82	1.19	51.66
S3b: excluding the 3 highest-mass-loss tests	M2	704	11	6.01	4.8	6.49	-0.8	0.34	46.26

Table S9. Programme metadata: steel origin, number of specimens with retained strength above 100 % (all retained in the analysis), number with reported ultimate strain (not used in any model) and number coded "Pre-loaded" in the database load-history field (exposed to structural testing before extraction; a blank field is not evidence of no pre-loading). *Database label "This Study (2023)".

Programme	n	Steel origin	Specimens with R > 100 %	Specimens with ultimate strain	Tests coded pre-loaded
Almusallam (2001)	42	Saudi Arabia	1	11	0
Cobo et al. (2011)	85	Spain	0	83	0
Fernandez et al. (2015)	35	Spain	0	35	35
Imperatore et al. (2017)	65	Italy	0	65	0
Kashani et al. (2013b)	5	United Kingdom	0	5	0
Meda et al. (2014)	22	Italy	0	22	16
Ou et al. (2016)	29	Taiwan	0	29	0
Tariq and Bhargava (2021)	120	India	1	120	0
Matthews et al. (2023)*	272	New Zealand	1	270	248
Vanama and Ramakrishnan (2020)	12	India	0	0	0
Zhang et al. (2012)	20	China	1	6	0
All (pooled specimens)	707	8 countries	4	646	299

Table S10. Programme-level summary of held-out LOPO errors (pp; error = predicted − observed; 11 programmes, 707 specimens). For M1–M3 each programme's predictions come from a different outer-fold fitted model, so the dispersion of programme bias includes variation among fitted models; it is not an identified variance component or a distribution for reliability analysis. The +5 pp threshold is exploratory.

Statistic	**M0**	**M1**	**M2**	**M3 (secondary)**
Equal-programme mean bias	+7.65	+1.29	+0.28	+0.17
Specimen-weighted mean bias	+5.17	−0.90	−0.85	−1.23
SD of programme bias	5.34	5.91	4.61	4.70
Minimum programme bias	+0.84	−5.36	−5.50	−5.54
Maximum programme bias	+16.20	+9.90	+6.75	+8.91
Range of programme bias	15.36	15.27	12.25	14.45
IQR of programme bias	7.99	9.34	7.60	5.05
Pooled within-programme residual SD	5.41	5.31	5.62	5.86
Total specimen-level residual SD	7.05	7.03	6.46	6.63
Programmes with bias > 0 (count)	11	6	4	4
Programmes with bias > +5 pp (count; exploratory)	6	3	3	2
Specimen-weighted SD of programme bias (secondary)	5.13	5.23	3.64	3.57

Table S11. Sensitivity of the programme-bias summary (pp): exclusion of Kashani et al. (2013b) (n = 5; 10 programmes) and leave-one-programme-out range of the SD of programme bias.

Analysis	**M0**	**M1**	**M2**	**M3 (secondary)**
Excluding Kashani: equal-programme mean bias	+6.90	+0.43	−0.35	−0.20
Excluding Kashani: SD of programme bias	4.98	5.46	4.32	4.78
Excluding Kashani: range of programme bias	15.36	15.27	12.25	14.45
Excluding Kashani: pooled within-programme SD	5.41	5.30	5.60	5.84
Leave-one-programme-out: SD of programme bias, min–max	4.77–5.63	5.46–6.23	4.31–4.85	3.89–4.95

Table S12. Models and validation designs. Tuning uses the inner grouped four-fold validation described in the main text, with the inner programme-macro MAE as criterion; m: average mass loss; d: nominal bar diameter.

(a) Models

Model	Form	Constraint or weighting	Tuned setting	Role
M0	$\hat{R} = 100 - m$; no fitted parameters	Uniform section loss	–	Physical benchmark; primary comparator
M1	Ridge regression on standardised m; fitted β_0, β_1	Intercept unpenalised; standardisation from training data	Penalty $\alpha \in \{0.01, 0.1, 1, 10, 100\}$	Primary
M2	Weighted isotonic regression; fitted step function	Non-increasing in m; equal programme weights; clipped outside the training range	–	Primary
M3	Histogram-based gradient boosting with inputs m and d	Non-increasing in m; no early stopping; fixed seed	Three configurations[a]	Secondary

(b) Validation designs

Design	Held-out unit	Folds	Tuning	Use
LOPO	Whole programme	11	Inner grouped 4-fold over the 10 training programmes	Primary evaluation; intervals and novelty from training programmes only
Record-random	Random specimens	5	Same nested grouped tuning	Diagnostic of the validation design only

[a](100 iterations, learning rate 0.05, maximum depth 2, minimum leaf size 20); (200, 0.05, 3, 20); (100, 0.10, 2, 40). Record-random seed 20261027.

Table S13. Overall LOPO performance and reliability. Programme-macro values are unweighted means over the 11 programmes; specimen-micro values pool all 707 specimens. PB: 95 % percentile interval from 5000 programme-level bootstrap resamples (descriptive). Mean positive overprediction is the mean of $\max(e, 0)$ per programme. Coverage ranges in brackets are minimum–maximum over programmes. n.c.: not computed (M3 is secondary and has no interval procedure).

Metric	M0	M1	M2	M3 (secondary)
Accuracy				
Programme-macro MAE (pp) [95% PB] – primary	**8.34 [5.68, 11.15]**	**6.76 [5.35, 8.13]**	**6.02 [4.80, 7.28]**	**6.34 [5.00, 7.65]**
Median programme MAE (pp) [IQR]	7.42 [4.61, 12.06]	5.62 [5.35, 8.85]	6.39 [4.48, 7.27]	6.40 [4.49, 8.28]
Specimen-micro MAE (pp)	5.86	5.68	4.83	5.04
RMSE (pp)	8.74	7.08	6.51	6.74
Bias				
Programme-macro bias (pp) [95% PB]	7.65 [4.78, 10.72]	1.29 [−1.81, 4.72]	0.28 [−2.14, 2.90]	0.17 [−2.23, 2.88]
Specimen-micro bias (pp)	5.17	−0.90	−0.85	−1.23
Non-conservative prediction				
Overpredicted, macro (%) [95% PB]	85.0 [75.4, 94.0]	52.3 [31.7, 73.8]	46.2 [30.0, 63.0]	44.3 [27.2, 63.0]
Overpredicted, micro (%)	80.6	37.2	35.6	30.7
Mean positive overprediction, macro (pp)	7.99	4.03	3.15	3.26
Exploratory 90% prediction intervals				
Coverage, specimen (%)	94.6	95.3	96.5	n.c.
Coverage, equal-programme (%) [min–max]	88.9 [60.0–100.0]	88.4 [60.0–100.0]	91.9 [70.0–100.0]	n.c.
Median width (pp)	38.5	27.7	31.2	n.c.

Table S14. Held-out performance for each of the 11 experimental programmes under leave-one-programme-out validation. Each row is one outer fold, in which the named programme was excluded from fitting, tuning and calibration. Bias is the mean of predicted − observed retained strength; positive values are non-conservative. Overpredicted: percentage of the programme's specimens with predicted > observed. Bold: lowest MAE of the three core models in the row (observed value; no significance implied). M3 and further metrics are given in Table S2.

		MAE (pp)			Signed bias (pp)			Overpredicted (%)		
Held-out programme	*n*	M0	M1	M2	M0	M1	M2	M0	M1	M2
Almusallam (2001)	42	**7.42**	9.26	7.77	+5.63	−2.48	−0.74	66.7	38.1	47.6
Cobo et al. (2011)	85	4.36	**2.75**	2.88	+4.03	+0.05	−0.59	89.4	45.9	47.1
Fernandez et al. (2015)	35	13.28	8.43	**6.76**	+13.28	+8.43	+6.75	100.0	100.0	97.1
Imperatore et al. (2017)	65	8.73	**5.62**	6.47	+7.98	+1.67	−0.60	89.2	55.4	40.0
Kashani et al. (2013b)	5	15.14	9.88	**8.02**	+15.14	+9.88	+6.67	100.0	100.0	60.0
Meda et al. (2014)	22	16.20	9.90	**6.39**	+16.20	+9.90	+6.22	100.0	100.0	81.8
Ou et al. (2016)	29	**3.51**	4.27	5.69	+2.25	−3.03	−5.13	69.0	27.6	20.7
Tariq & Bhargava (2021)	120	10.85	**5.34**	5.51	+10.32	+3.84	+2.51	97.5	76.7	70.0
Matthews et al. (2023)*	272	**1.54**	5.43	3.45	+0.84	−5.31	−3.29	68.4	1.5	5.1
Vanama & Ramakrishnan (2020)	12	4.86	5.36	**3.21**	+4.86	−5.36	−3.19	100.0	0.0	8.3
Zhang et al. (2012)	20	**5.84**	8.12	10.03	+3.59	−3.38	−5.50	55.0	30.0	30.0

Supplementary figures

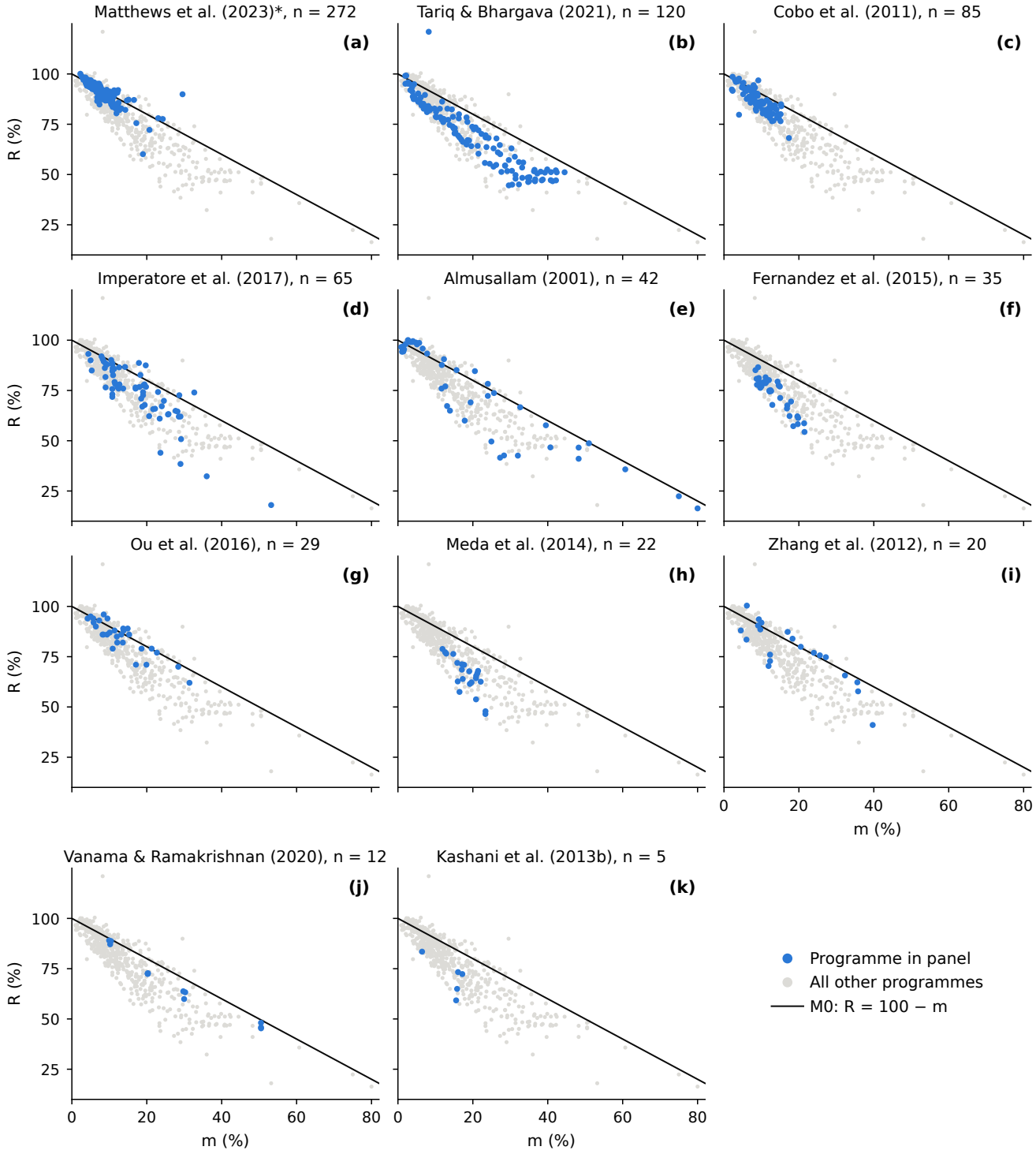


Fig. S1. Retained ultimate strength R versus average mass loss m for each of the 11 experimental programmes (blue), shown against all other programmes (grey), with the uniform-section-loss line $R = 100 - m$. Panels are ordered by number of tests (n).

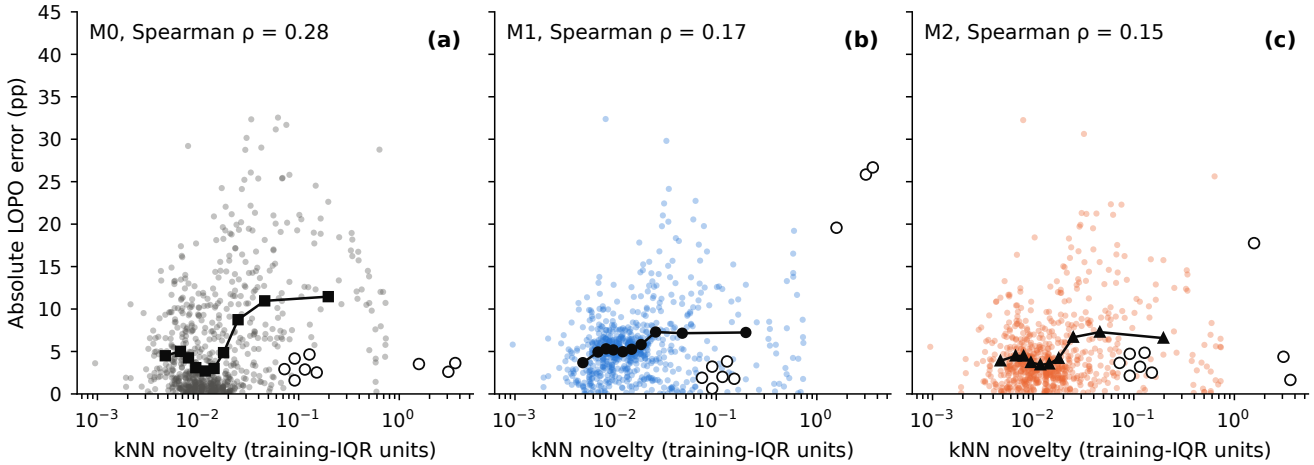


Fig. S2. Absolute LOPO error versus k-nearest-neighbour mass-loss novelty (mean distance to the 10 nearest training values, in training-IQR units; logarithmic axis) for (a) M0, (b) M1 and (c) M2, with decile means (black line) and pooled Spearman ρ (descriptive). Open circles: the 9 specimens outside the training mass-loss range.

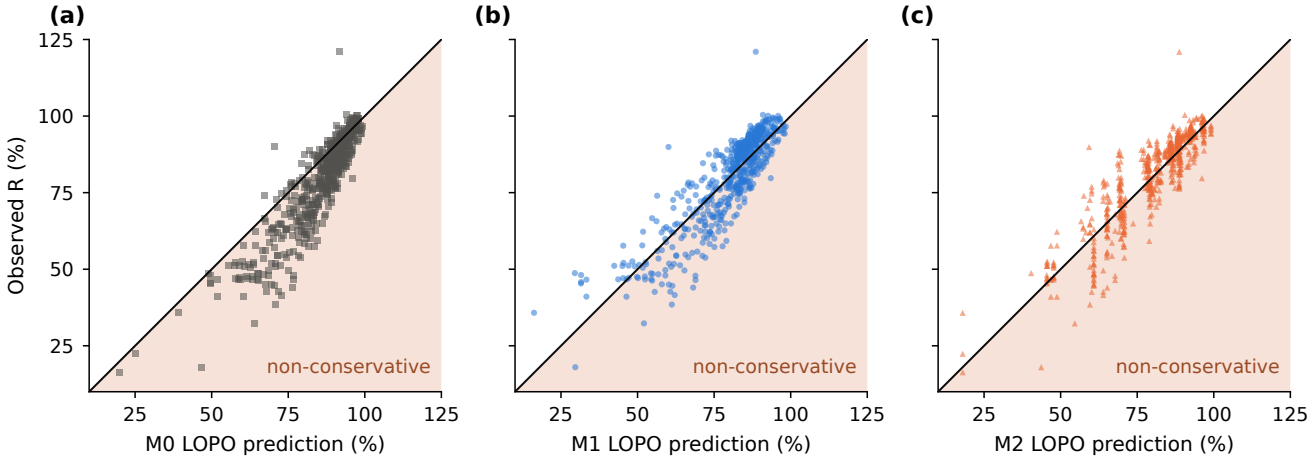


Fig. S3. Observed versus LOPO-predicted retained strength for all 707 specimens under (a) M0, (b) M1 and (c) M2, with the identity line. The tint marks non-conservative predictions (predicted > observed).

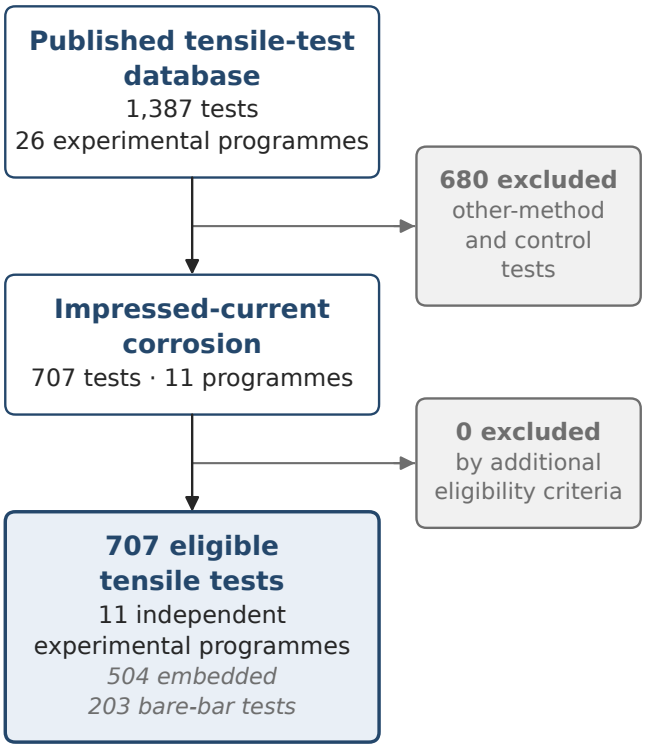


Fig. S4. Selection of the analysis cohort from the published tensile-test database (Matthews et al. 2024b). Of 1,387 tests from 26 experimental programmes, 707 tests from 11 programmes used impressed-current corrosion. All 707 met the additional prespecified eligibility criteria. The final cohort comprises 504 embedded and 203 bare-bar tests. Diagram prepared with Python (Matplotlib) code written with the assistance of Claude (Anthropic); counts computed from the source database.

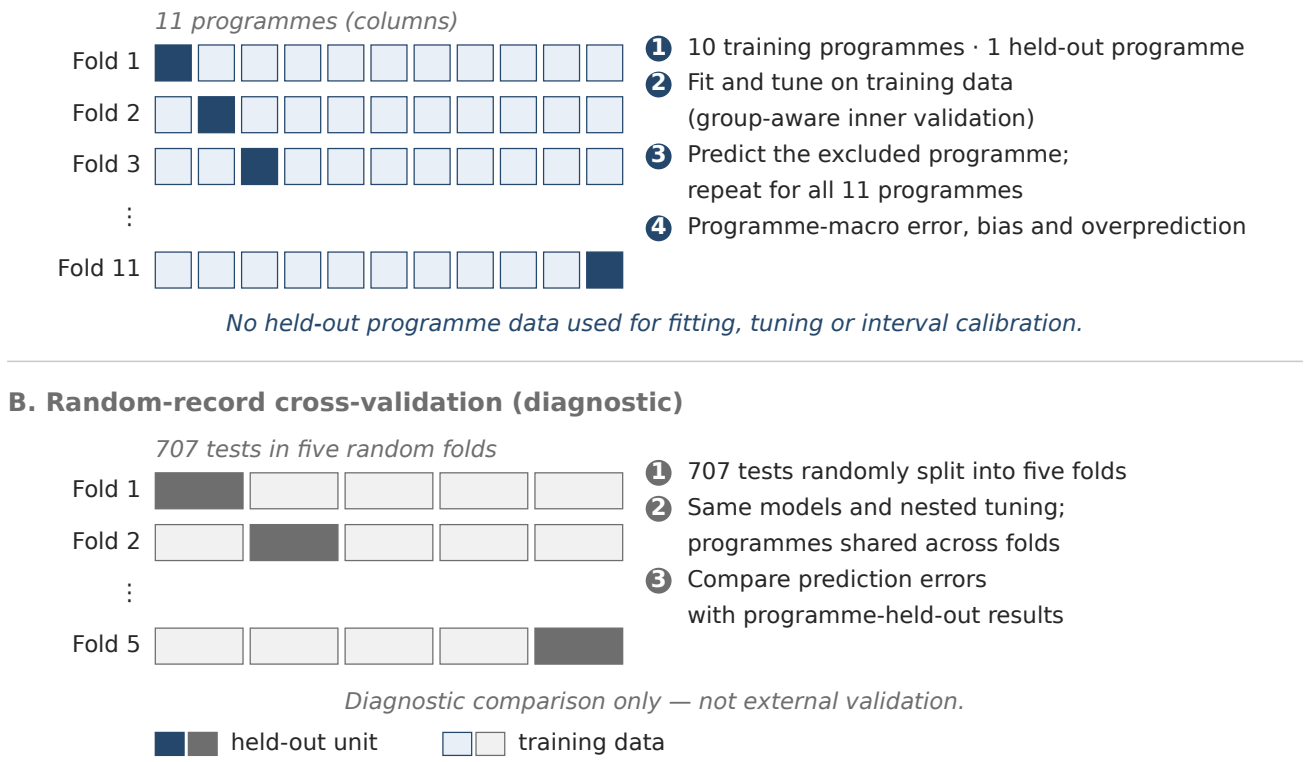


Fig. S5. Comparison of the two validation strategies. (A) Leave-one-programme-out validation holds out each of the 11 experimental programmes in turn; model fitting, grouped hyperparameter tuning and interval calibration use training programmes only. (B) Random-record five-fold cross-validation provides a diagnostic comparison using the same model families. Primary reliability metrics are calculated from held-out predictions and summarised at programme level. Rows represent schematic validation folds; shaded cells indicate the held-out unit. The schematic is not proportional to programme size. Diagram prepared with Python (Matplotlib) code written with the assistance of Claude (Anthropic); it depicts the procedure implemented in the analysis code.

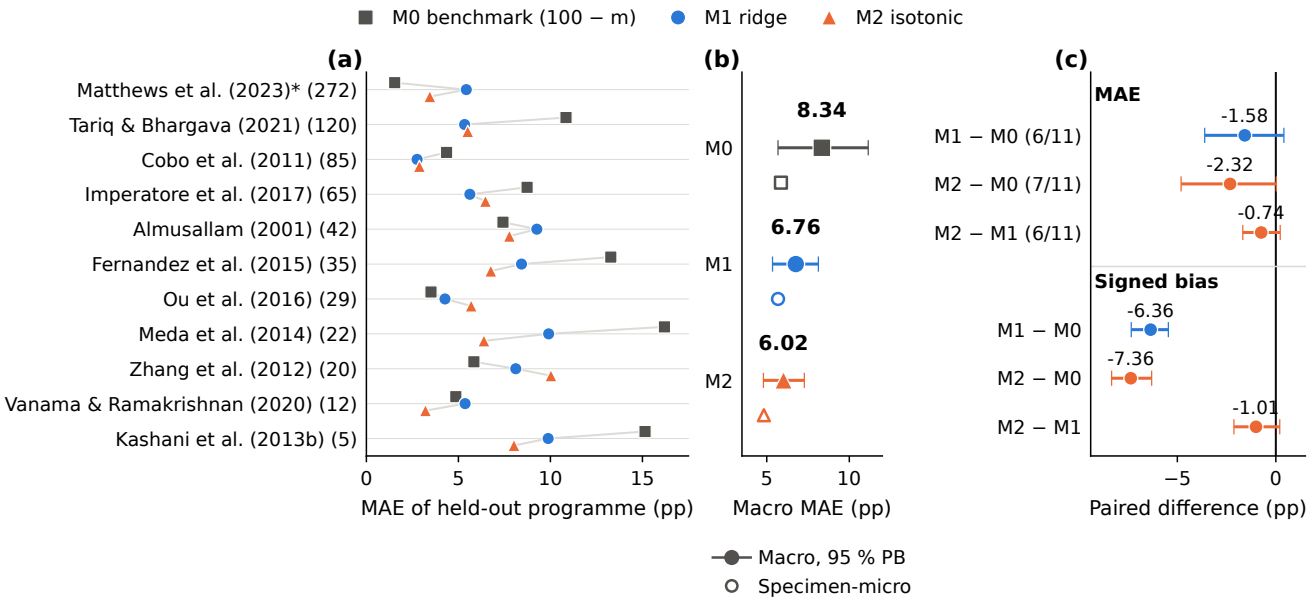


Fig. S6. Predictive accuracy under leave-one-programme-out (LOPO) validation. (a) Mean absolute error (MAE) of each held-out programme for M0, M1 and M2; grey lines join the models within a programme. (b) Programme-macro MAE, the primary metric, which weights the 11 programmes equally (filled symbols, labelled; 95 % percentile intervals from 5000 programme-level bootstrap resamples, PB), and specimen-micro MAE (open symbols). (c) Paired programme-bootstrap differences in programme-macro MAE and in programme-macro signed bias (95 % PB); the number of programmes in which the first model had the lower MAE is given in parentheses. Intervals that include zero do not establish a difference; all intervals are descriptive.

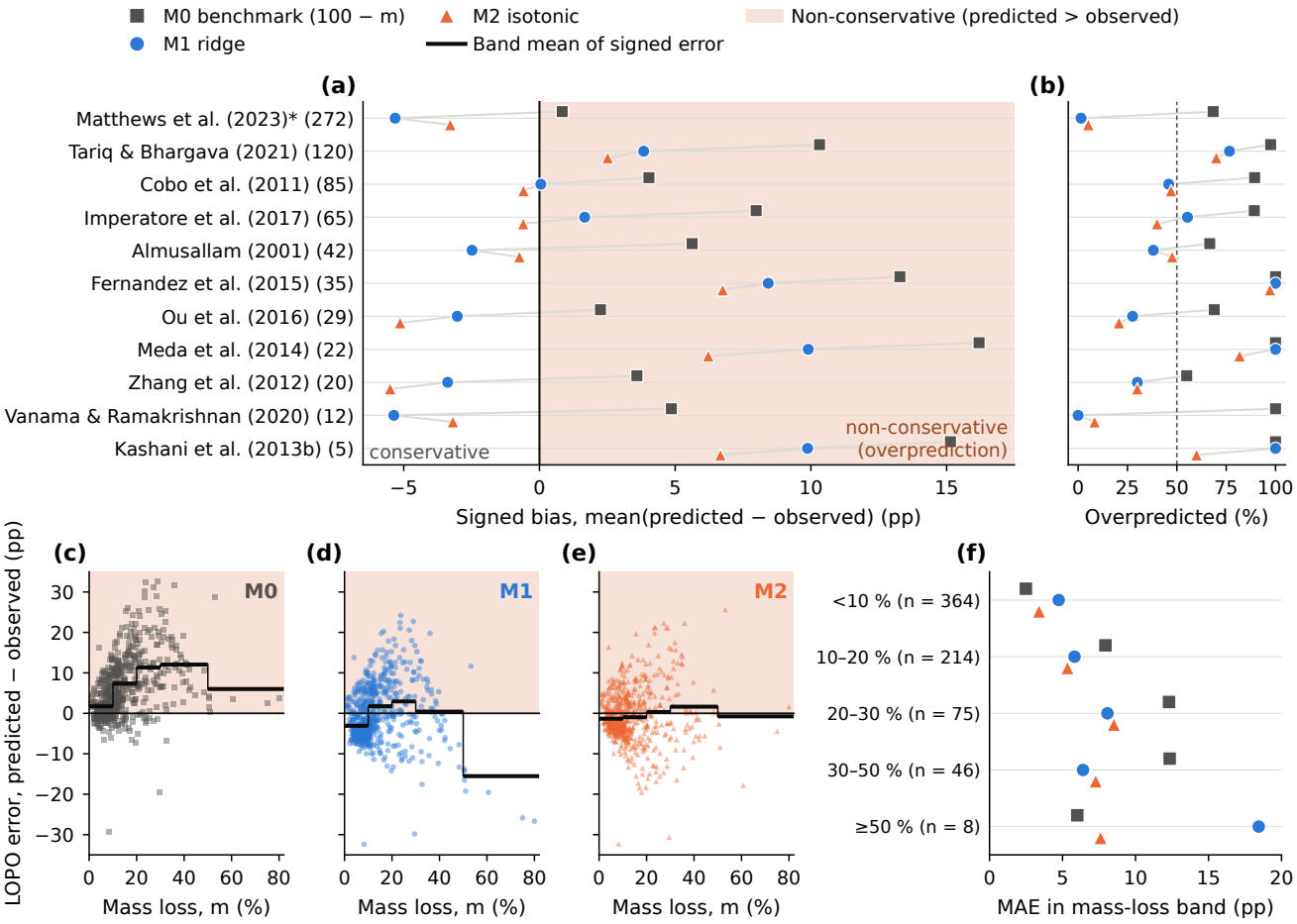


Fig. S7. Direction of held-out error under LOPO. The tint marks non-conservative error (prediction above observation). (a) Signed bias, mean(predicted − observed), and (b) percentage of specimens overpredicted, for each held-out programme; dashed line in (b): 50 %. (c–e) Signed error against mass loss for every specimen under M0, M1 and M2, with the mean signed error in each predeclared mass-loss band (black steps). (f) MAE in each mass-loss band. The 30–50 % and ≥50 % bands contain 46 specimens from 6 programmes and 8 specimens from 3 programmes (Table S5). Observed versus predicted values are shown in Fig. S3.

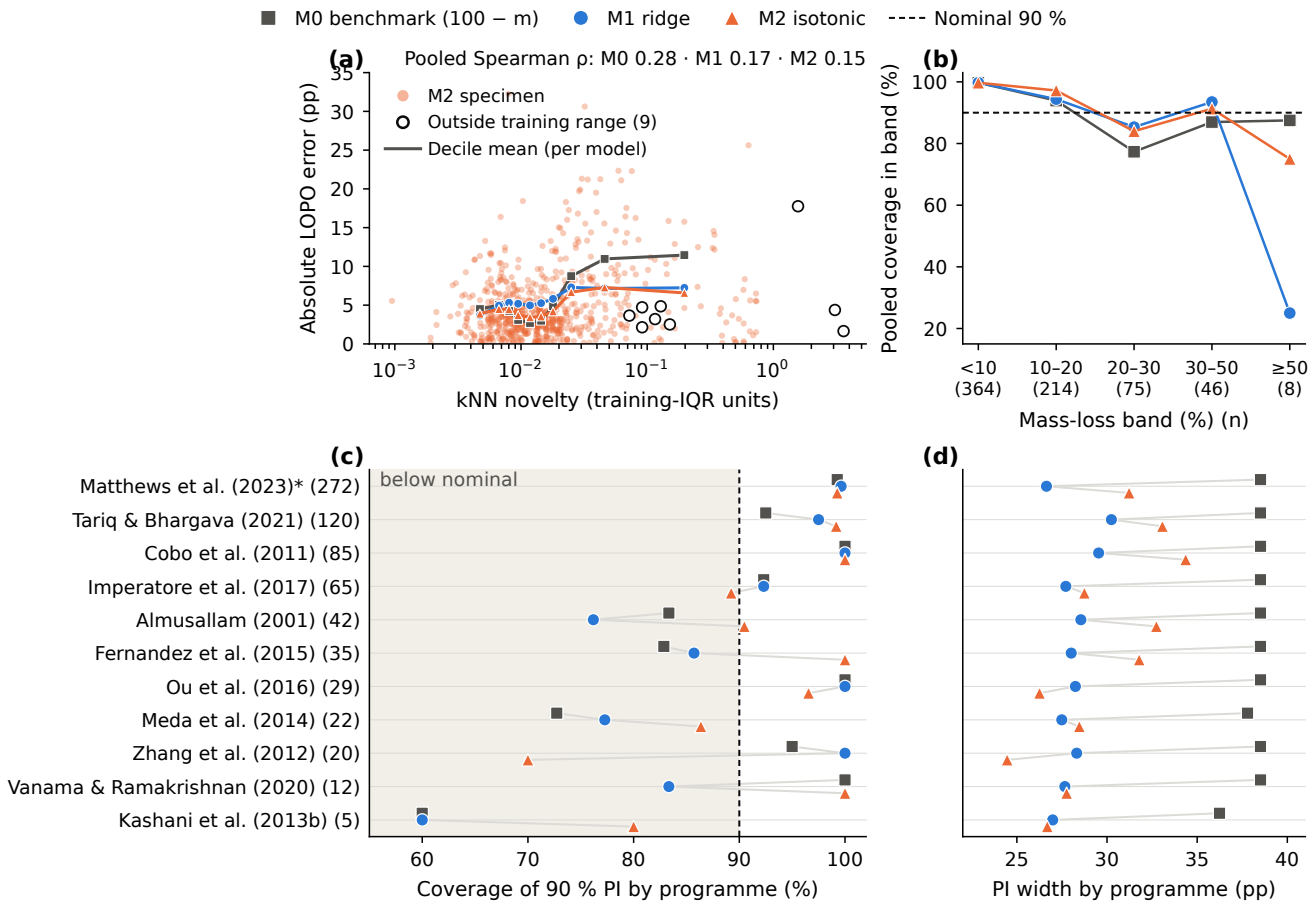


Fig. S8. Training-domain novelty and exploratory 90 % prediction intervals (PIs) under LOPO. (a) Absolute error versus k-nearest-neighbour novelty (mean distance to the 10 nearest training mass-loss values, in training-IQR units; logarithmic axis) for M2 specimens, with decile means for M0, M1 and M2 and pooled Spearman ρ (descriptive); open circles: the 9 specimens, all from Almusallam (2001), outside the training mass-loss range (M0 and M1 scatters in Fig. S2). (b) Pooled PI coverage in each mass-loss band (n in parentheses). (c) PI coverage and (d) PI width ($2q$) for each held-out programme; shading in (c): below nominal. Kashani et al. (2013b) has 5 tests, so its coverage changes in steps of 20 percentage points. Half-widths were calibrated in each fold on residuals of the 10 training programmes with equal programme weight; because held-out programmes are shifted, no coverage guarantee applies.